COMP4141

Theory of Computation

Lecture 7: Properties of CFLs

Administrivia

- Formatif Task 2 now visible
  - P/C/D tasks: show your tutor your solution this week;
  - HD task: due next week (formal proof) submit pdf
- Formatif Task 3 up tonight – show your tutor this week or next.
- Remaining tasks up asap (well before census)
- Formatif Task 1 feedback to be completed by Friday
- Air conditioning issue being addressed

Outline

Pumping Lemma for CFLs

Examples of Non-Context-Free Languages

Closure Properties of CFLs Revisited

Overnight problem

How to pump CFLs

Similar to regular languages, CFLs can be pumped.

Theorem (Pumping Lemma)

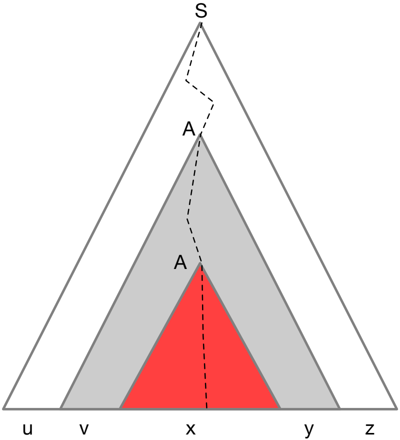
If $L \subseteq \Sigma^*$ is context-free then there exists $p \in \mathbb{N}$ (the pumping length) where, if $w \in L$ with $|w| \geq p$, then w may be split into five pieces, $w = uvxyz$, satisfying the following conditions:

1. $|vy| > 0$, *and*
2. $|vxy| \leq p$.
3. $uv^ixy^iz \in L$, *for all* $i \in \mathbb{N}$,

Pumping Lemma proof

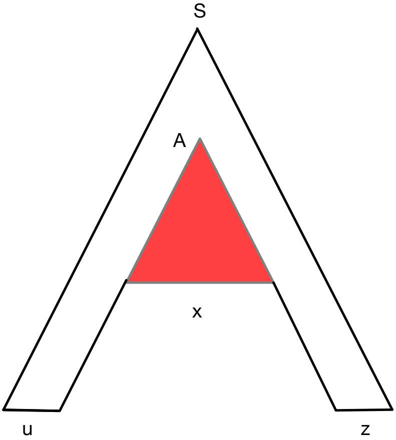
The parse tree of a CFG has bounded branching, so the parse tree for a long word has a long branch.

The CFG has a finite set of nonterminals, so any sufficiently long branch has a repeated nonterminal A (low down in the tree, so vxy is short.



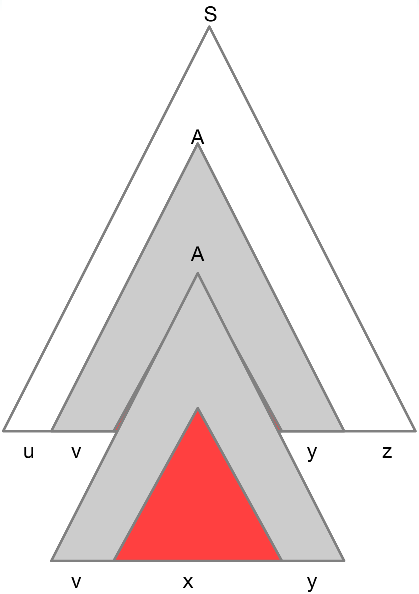
Pumping Lemma proof

Pumping down yields uxz



Pumping Lemma proof

Pumping up yields uv^2xy^2z. Repeating i times yields $uv^{i+1}xy^{i+1}z$.



Outline

A Non-Context-Free Language

Example

Let $L_1 = \{\ a^i b^i c^i \mid i > 0\ \}$.

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Let p be G's pumping length. Consider $w = \mathrm{a}^p\mathrm{b}^p\mathrm{c}^p$ and let $w = uvxyz$ satisfy the conditions of the pumping lemma.

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If vxy is in $\mathtt{a}^*\mathtt{b}^*$ then uxz is not in $L(G_1)$ because, by condition 1 vy contains at least one symbol, so uxz has either fewer than p copies of a or b but exactly p copies of c.

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If vxy is in $\mathtt{b}^*\mathtt{c}^*$ we reason analogously.

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If vxy is in $\mathtt{b}^*\mathtt{c}^*$ we reason analogously.

Due to condition 2 of the pumping lemma there are no other cases. Each case lead to a contradiction. Thus $L(G_1)$ cannot be context-free.

Another Non-Context-Free Language

Example

Let $L_2 = \{\ ww \mid w \in \{0,1\}^*\ \}$
Assume that L_2 is a CFL and let p be the pumping length for L_2.

Attempt 1: Take $w = 0^p 1 0^p 1$...

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As a rule of thumb, we suggest words w based on p such that only a small number of cases for partitioning w into $uvxyz$ satisfying $|vxy| \leq p$ and $vy \neq \epsilon$ emerge. This is often achieved by making all the different regions in w sufficiently long such that vxy must be located in at most two of them.

Another Non-Context-Free Language

Example

Let $L_2 = \{\ ww \mid w \in \{0,1\}^*\ \}$
Assume that L_2 is a CFL and let p be the pumping length for L_2.

Attempt 2: Take $w = 0^p 1^p 0^p 1^p$:

As in the failed attempt, w contains four regions, but this time vxy can range over at most two of them.

Another Non-Context-Free Language

Example

Let $L_2 = \{\ ww \mid w \in \{0,1\}^*\ \}$
Assume that L_2 is a CFL and let p be the pumping length for L_2.

Attempt 2: Take $w = 0^p1^p0^p1^p$:

If vxy is located in a single one of the regions, i.e., $vxy \in 0^* \cup 1^*$ pumping either way takes us out of L.

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Example

Let $L_2 = \{\ ww \mid w \in \{0,1\}^*\ \}$
Assume that L_2 is a CFL and let p be the pumping length for L_2.

Attempt 2: Take $w = 0^p1^p0^p1^p$:

Otherwise, if vxy is stretches across two adjacent regions we distinguish three cases:

1. $u \in 0^*$ and $vxy \in 0^*1^*$, that is, vxy doesn't straddle the midpoint of w but is contained in the first half of w. Pumping down leads to a word the midpoint of which is located inside the third region, 0^p. So the left half of uxz ends in a 0 but the right half ends in a 1. Thus $uxz \notin L$.
2. $u \in 0^p1^p0^*$ and $vxy \in 0^*1^*$, that is, vxy is contained in the second half of w. This case is similar to the previous one.
3. $vxy \in 1^*0^*$, that is, vxy straddles the midpoint of w. Pumping down leads to $uxz = 0^p1^i0^j1^p$ for some $i, j \le p$ that cannot both be p. Again, $uxz \notin L$. $\square$

Outline

Pumping Lemma for CFLs

Examples of Non-Context-Free Languages

Closure Properties of CFLs Revisited

Overnight problem

Closure under Union

Theorem

CFLs are closed under union.

Proof.

Let $L_1, L_2 \subseteq \Sigma^*$ be context-free.

For $i = 1, 2$ let $G_i = (V_i, \Sigma, P_i, S_i)$ be a CFG such that $L(G_i) = L_i$ such that w.l.o.g. $V_1 \cap V_2 = \emptyset$.

Consider $G = (V_1 \cup V_2 \cup \{S\}, \Sigma, P, S)$ for some fresh non-terminal $S \notin V_1 \cup V_2 \cup \Sigma$ where

$$P = P_1 \cup P_2 \cup \{S \to S_1 \mid S_2\}$$

This grammar clearly generates $L_1 \cup L_2$. □

Closure under Concatenation

Theorem

CFLs are closed under concatenation.

Proof.

With L_1, L_2, G_1, G_2, G as in the previous proof, except for

$$P = P_1 \cup P_2 \cup \{S \to S_1 S_2\}$$

This grammar clearly generates $L_1 L_2$. $\square$

Closure under Kleene star

Theorem

CFLs are closed under Kleene star (*).

Proof: exercise.

Intersection

Theorem

CFLs are not closed under intersection.

A counterexample suffices. We introduce two languages

$$\begin{aligned} L_1 &= \{\, a^i b^i c^j \mid i > 0 \land j > 0 \,\} \\ L_2 &= \{\, a^j b^i c^i \mid i > 0 \land j > 0 \,\} \end{aligned}$$

which are context-free but whose intersection

$$L_1 \cap L_2 = \{\, a^i b^i c^i \mid i > 0 \,\}$$

is not as we've shown. It remains to be shown that the L_i are indeed CFLs.

Intersection

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Consider the two CFGs

$$\begin{aligned} G_1 : S &\to AB \\ A &\to aAb \mid ab \\ B &\to cB \mid c \end{aligned} \qquad \begin{aligned} G_2 : S &\to AB \\ A &\to aA \mid a \\ B &\to bBc \mid bc \end{aligned}$$

In G_1, the non-terminal A generates all strings of the form $a^i b^i$, and B generates all strings of cs.

In G_2, the non-terminal B generates all strings of the form $b^i c^i$, and A generates all strings of as. □

Complementation

That CFLs are not closed under complementation follows from de Morgan's laws:

$$L_1 \cap L_2 = \overline{\overline{L_1} \cup \overline{L_2}} \ .$$

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Concrete example

$$L = \{a, b, c\}^* \setminus \{a^i b^i c^i : i \in \mathbb{N}\}$$

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$$\begin{aligned} L &= \{a, b, c\}^* \setminus \{a^i b^i c^i : i \in \mathbb{N}\} \\ &= \{w : w \text{ not of the form } a^* b^* c^*\} \\ &\quad \cup \quad \{a^i b^j c^k : j \neq k\} \quad \cup \quad \{a^i b^j c^k : j \neq i\} \end{aligned}$$

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Non-determinism

Non-closure under complementation tells us that, unlike NFAs, PDAs accept *strictly more* languages than their deterministic counterparts.

$$\textsc{Regular} \subsetneq \textsc{Deterministic PDAs} \subsetneq \textsc{Context-Free}$$

Other closure properties

- The intersection of a CFL and a regular language is a CFL
- CFLs are closed under homomorphisms and inverse homomorphisms
- CFLs are closed under $\mathrm{rev}(\cdot)$ and $\mathrm{odd}(\cdot)$ (Why?)

Outline

Overnight problem

Question

Given a word w and a CFG G, how can we tell if $w \notin L(G)$?